

Examination Solutions

Database Management System (E1PA104B)

Course: Masters of Computer Applications (MCA) [cite: 28]

Examination: Mid Term - Nov 2025 [cite: 27]

School: School of Computer Applications and Technology [cite: 26]

1. Overcoming File System Issues

DBMS provides better data handling than traditional file systems by addressing: [cite: 32, 33]

- **Data Redundancy/Inconsistency:** Centralized storage prevents multiple copies of the same data.
- **Data Isolation:** Allows sharing data across different applications easily.
- **Integrity Problems:** Enforces constraints (like non-negative balances).
- **Atomicity:** Ensures transactions are "all or nothing," preventing partial updates.
- **Concurrent Access:** Manages multiple users accessing the data simultaneously without conflicts.

2. Overall Structure of a Database System

The structure consists of: [cite: 34]

- **Users:** Naive users, application programmers, sophisticated users, and DBAs.
- **Query Processor:** DDL interpreter, DML compiler, and Query evaluation engine.
- **Storage Manager:** Buffer manager, File manager, Authorization and integrity manager.
- **Disk Storage:** Data files, Data dictionary, Indices, and Statistical data.

3. Database Languages for a School System

For updating and summarizing marks, two primary languages are used: [cite: 37, 38]

- **DML (Data Manipulation Language):** Used for updating records.
Syntax: `UPDATE table_name SET column = value WHERE condition;`
- **DQL (Data Query Language):** Used for summarizing (SELECT with Aggregate functions).
Example: `SELECT AVG(marks) FROM Student_Marks;`

4. Three-Level Architecture

This architecture separates the physical database from user applications: [cite: 40, 41]

- **External/View Level:** Highest level, describing how individual users see the data.
- **Conceptual/Logical Level:** Describes what data is stored and the relationships between them.
- **Internal/Physical Level:** Lowest level, describing how data is actually stored on storage media.

5. Role of Database Schema in Retail

The schema serves as the logical blueprint for the retail store: [cite: 41]

- Defines **tables** for Products, Sales, and Inventory.
- Establishes **relationships** (e.g., Sales table links to Product ID).
- Enforces **Data Integrity** to ensure prices and stock counts are accurate.

6. E-R Diagram Components (E-Learning)

The diagram for an e-learning platform includes: [cite: 42]

- **Entities:** Student, Course, Instructor.
- **Relationships:** Students *enroll* in Courses; Instructors *teach* Courses.
- **Attributes:** Student (ID, Name), Course (Code, Title), Instructor (EmpID, Specialization).

7. Data Independence

Achieved via mappings between the three levels: [cite: 43]

- **Logical Data Independence:** Ability to change the conceptual schema without changing external schemas (e.g., adding a new column).
- **Physical Data Independence:** Ability to change the physical storage structures (e.g., switching from B-trees to Hashing) without changing the conceptual schema.

8. Mapping E-R to Relational

Mapping converts entities and relationships into tables: [cite: 49, 50]

- Strong entities become tables with the entity's ID as the **Primary Key**.

- 1:N relationships result in the primary key of the '1' side becoming a **Foreign Key** in the 'N' side.

9. Bank Transaction Security (DCL & TCL)

Justification for using these commands in banking: [cite: 51, 52, 53]

- **DCL (Data Control Language):** Uses **GRANT** and **REVOKE** to ensure only authorized staff access sensitive financial data.
- **TCL (Transaction Control Language):** Uses **COMMIT** and **ROLLBACK** to ensure that if a fund transfer fails halfway, the data returns to its original consistent state.